

# TITAN research-software demonstration

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## Research-software output: internally derived TCGA estimates

### COAD - TCGA-AA-3972

**UNVALIDATED RESEARCH OUTPUT <U+2014> BINARY TCGA OOF SCORE RANK (NOT PROBABILITY) <U+2014> NO INDEPENDENT EXTERNAL VALIDATION <U+2014> NOT FOR DIAGNOSIS, TREATMENT SELECTION OR CLINICAL RISK ESTIMATION**

| Models applied | Continuous | Binary | Slides pooled |
|----------------|------------|--------|---------------|
| 19             | 10         | 9      | 1             |

This research-software demonstration applies models developed and evaluated only in TCGA. Every displayed value is an **internally derived TCGA estimate**, not external performance evidence or a diagnostic output. Binary LDA scores are uncalibrated. Every displayed **TCGA OOF score rank (not probability)** describes relative position only.

By default, TITANPred excludes binary models with fewer than 50 participants in either development class. Such models appear only after the caller explicitly sets `include_limited_evidence = TRUE`.

### Prominent tissue-source-site sensitivity warning

**12 model(s) in this output fell below their original prespecified screening threshold under TCGA tissue-source-site-grouped internal validation. These calls must not be highlighted without this warning.**

Tissue-source-site grouping is not institutional or scanner-level external validation.

| Endpoint (type)                                      | TSS-grouped metric | Grouped minus random | Warning category          |
|------------------------------------------------------|--------------------|----------------------|---------------------------|
| APC (binary)                                         | BA 0.543           | -0.250               | Near chance or worse      |
| PI3K (binary)                                        | BA 0.593           | -0.020               | Below prespecified screen |
| Aneuploidy score (continuous)                        | Q2 0.173           | -0.086               | Below prespecified screen |
| Deleted arm count (continuous)                       | Q2 0.147           | -0.066               | Below prespecified screen |
| MANTIS score (continuous)                            | Q2 0.113           | -0.104               | Below prespecified screen |
| MSIsensor score (continuous)                         | Q2 0.026           | -0.229               | Below prespecified screen |
| Aneuploidy Score (continuous)                        | Q2 0.174           | -0.096               | Below prespecified screen |
| Lymphocyte Infiltration Signature Score (continuous) | Q2 0.048           | -0.238               | Below prespecified screen |
| Nonsilent Mutation Rate (continuous)                 | Q2 0.088           | -0.127               | Below prespecified screen |
| SNV Neoantigens (continuous)                         | Q2 -0.013          | -0.252               | Near chance or worse      |
| Silent Mutation Rate (continuous)                    | Q2 -0.003          | -0.236               | Near chance or worse      |
| TIL Regional Fraction (continuous)                   | Q2 0.176           | -0.278               | Below prespecified screen |

Case context (not used as model input)

**Shared selection features.** Both cases were men with sigmoid-colon, moderately differentiated (G2), pT3 adenocarcinoma and tumour-free resection margins. Nodal status was not matched: example A was pN1, whereas the available slide summary for example B did not state a nodal category.

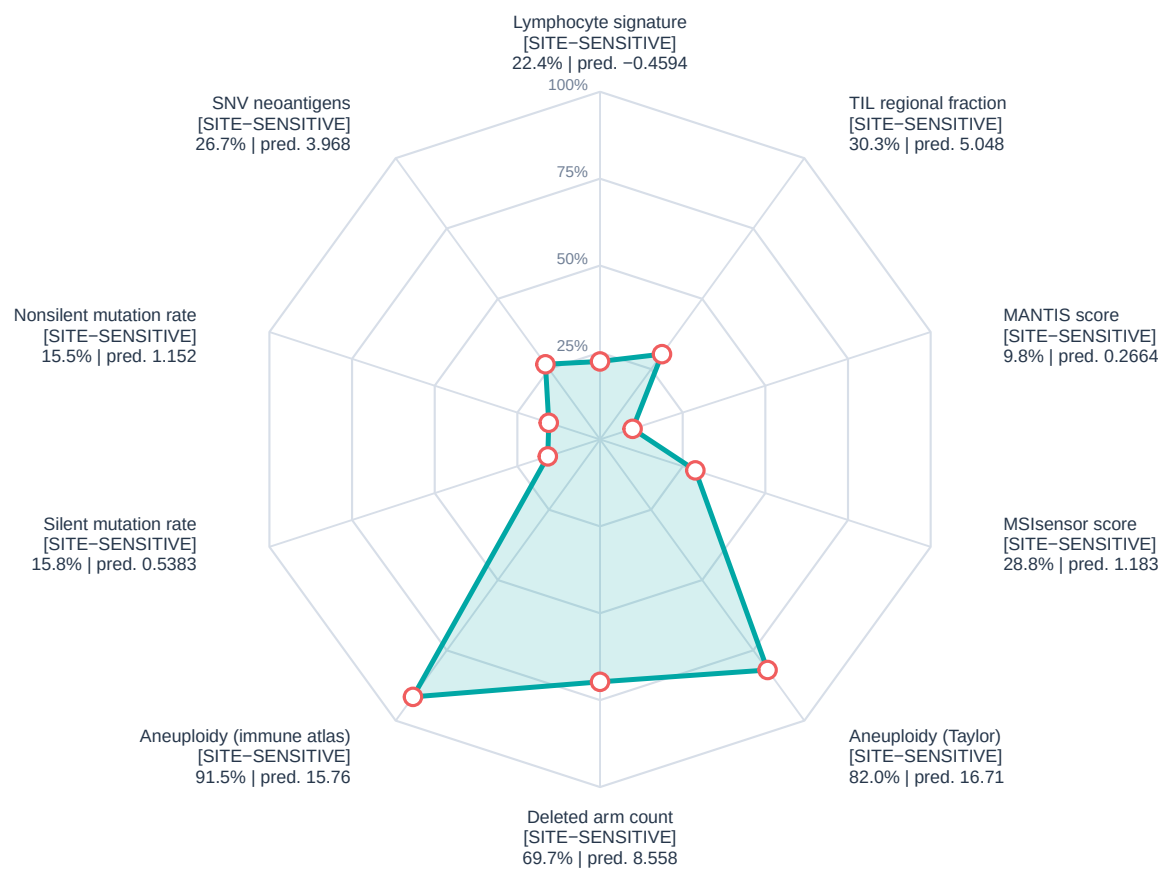
**Pathology summary.** The sigmoid colectomy contained an ulcerated grade-2 colorectal adenocarcinoma extending through the bowel wall into adjacent mesocolic fat (pT3). The proximal and distal resection margins were uninvolved; the supplied slide summary did not state a nodal category.

**Clinical-context source.** Pathology was paraphrased from TCGA-Slide-Reports.csv distributed with the TITAN study (Ding et al., Nature Medicine 2025; doi:10.1038/s41591-025-03982-3).

The pathology field is descriptive context only; it was not a predictor, training target, or validation outcome for TITANPred. Treatment and response narratives are intentionally excluded from this research-software output.

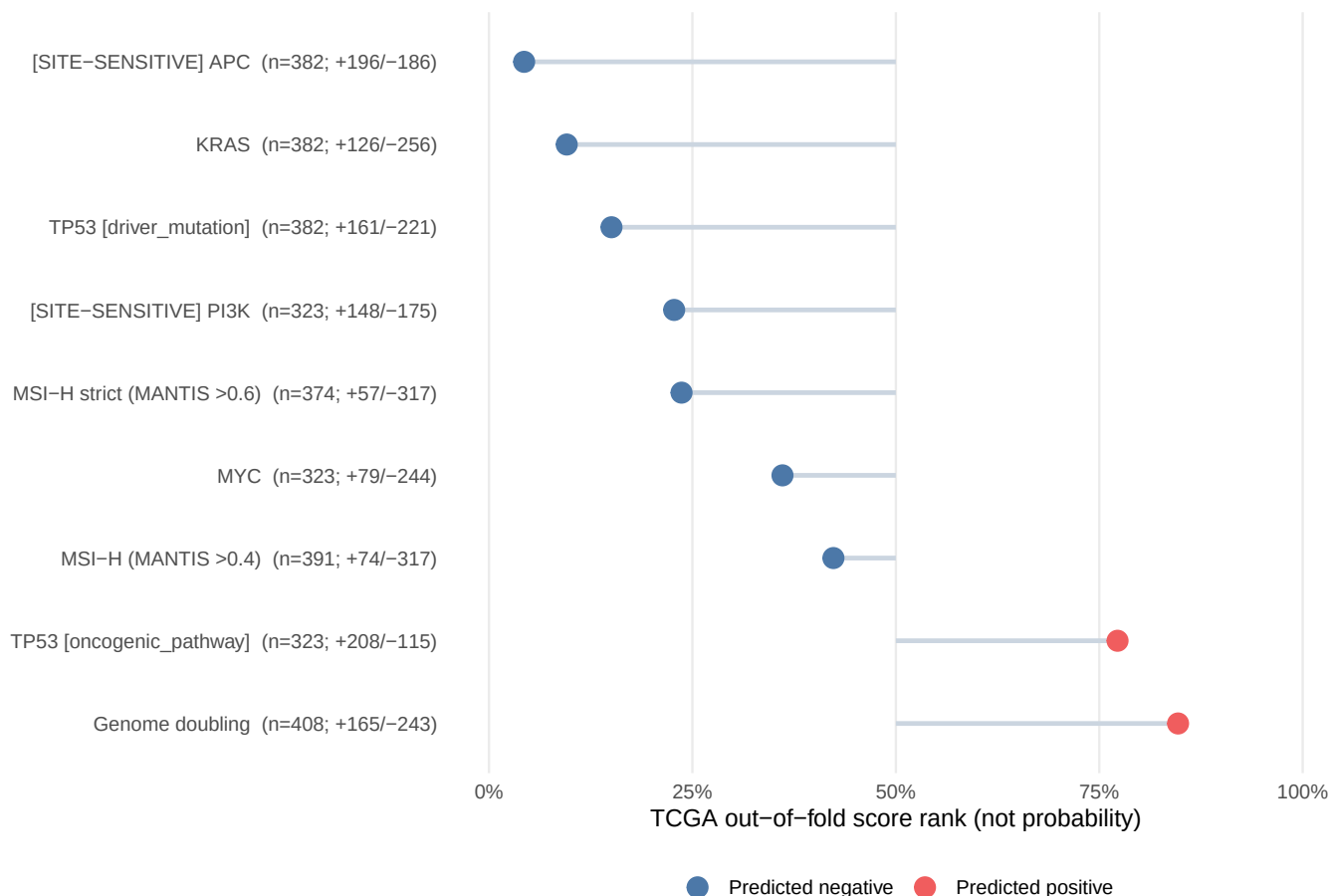
Continuous internally derived TCGA estimates

[SITE-SENSITIVE] marks models below the original prespecified screening threshold under TCGA TSS grouping



## Binary research-model calls – rank is not probability

TCGA out-of-fold LDA-score rank (not probability); [SITE-SENSITIVE] models req



## Continuous internally derived estimates

| Family                     | Endpoint                                | Prediction | TCGA OOF prediction percentile (not prob) |
|----------------------------|-----------------------------------------|------------|-------------------------------------------|
| thorsson                   | Aneuploidy Score                        | 15.76      | 91.5%                                     |
| microsatellite_instability | MANTIS score                            | 0.2664     | 9.8%                                      |
| thorsson                   | Nonsilent Mutation Rate                 | 1.152      | 15.5%                                     |
| thorsson                   | Silent Mutation Rate                    | 0.5383     | 15.8%                                     |
| aneuploidy                 | Aneuploidy score                        | 16.71      | 82.0%                                     |
| thorsson                   | Lymphocyte Infiltration Signature Score | -0.4594    | 22.4%                                     |
| thorsson                   | SNV Neoantigens                         | 3.968      | 26.7%                                     |
| microsatellite_instability | MSIsensor score                         | 1.183      | 28.8%                                     |
| aneuploidy                 | Deleted arm count                       | 8.558      | 69.7%                                     |
| thorsson                   | TIL Regional Fraction                   | 5.048      | 30.3%                                     |

**Internal development evidence.** The primary screen and five-repeat estimates use different nested-CV partitions and need not be identical.

| Endpoint                                | Primary screen Q2 | Five-repeat Q2 | Five-repeat RMSE | Five-repeat Spearman |
|-----------------------------------------|-------------------|----------------|------------------|----------------------|
| Aneuploidy score                        | 0.260             | 0.294          | 6.879            | 0.570                |
| Deleted arm count                       | 0.213             | 0.196          | 5.742            | 0.565                |
| MANTIS score                            | 0.217             | 0.243          | 0.201            | 0.392                |
| MSIsensor score                         | 0.254             | 0.218          | 9.707            | 0.269                |
| Aneuploidy Score                        | 0.269             | 0.272          | 5.936            | 0.555                |
| Lymphocyte Infiltration Signature Score | 0.285             | 0.285          | 0.726            | 0.532                |
| Nonsilent Mutation Rate                 | 0.214             | 0.247          | 0.805            | 0.335                |
| SNV Neoantigens                         | 0.239             | 0.208          | 0.969            | 0.368                |
| Silent Mutation Rate                    | 0.234             | 0.242          | 0.651            | 0.351                |
| TIL Regional Fraction                   | 0.454             | 0.423          | 6.428            | 0.688                |

## Binary research-model calls

### Calls and development denominators.

| Endpoint                   | Class | LDA score | TCGA OOF score rank (not probability) | Training n (+/-) |
|----------------------------|-------|-----------|---------------------------------------|------------------|
| Genome doubling            | 1     | 1.557     | 84.7%                                 | 408 (+165/-243)  |
| APC                        | 0     | -3.918    | 4.3%                                  | 382 (+196/-186)  |
| KRAS                       | 0     | -4.625    | 9.6%                                  | 382 (+126/-256)  |
| TP53 [driver_mutation]     | 0     | -3.37     | 15.1%                                 | 382 (+161/-221)  |
| MSI-H (MANTIS >0.4)        | 0     | -4.847    | 42.3%                                 | 391 (+74/-317)   |
| MSI-H strict (MANTIS >0.6) | 0     | -9.961    | 23.7%                                 | 374 (+57/-317)   |
| MYC                        | 0     | -2.331    | 36.1%                                 | 323 (+79/-244)   |
| PI3K                       | 0     | -1.268    | 22.8%                                 | 323 (+148/-175)  |
| TP53 [oncogenic_pathway]   | 1     | 1.99      | 77.2%                                 | 323 (+208/-115)  |

**Internal development evidence. Primary-screen BA and five-repeat estimates use different nested-CV partitions and need not be identical. PPV and NPV use the observed TCGA prevalence.**

| Endpoint                   | Primary screen BA | Five-repeat BA | Five-repeat AUROC | Five-repeat PR-AUC | PPV   |
|----------------------------|-------------------|----------------|-------------------|--------------------|-------|
| Genome doubling            | 0.691             | 0.692          | 0.760             | 0.678              | 0.636 |
| APC                        | 0.793             | 0.794          | 0.848             | 0.814              | 0.760 |
| KRAS                       | 0.680             | 0.692          | 0.781             | 0.581              | 0.600 |
| TP53 [driver_mutation]     | 0.779             | 0.778          | 0.854             | 0.793              | 0.718 |
| MSI-H (MANTIS >0.4)        | 0.744             | 0.732          | 0.834             | 0.634              | 0.677 |
| MSI-H strict (MANTIS >0.6) | 0.784             | 0.782          | 0.915             | 0.741              | 0.750 |
| MYC                        | 0.603             | 0.605          | 0.676             | 0.417              | 0.515 |
| PI3K                       | 0.613             | 0.607          | 0.640             | 0.583              | 0.582 |
| TP53 [oncogenic_pathway]   | 0.671             | 0.668          | 0.734             | 0.815              | 0.752 |

## Binary development reliability

### Fold class counts and component selection.

| Endpoint                   | Minimum outer test (+/-) | Minimum inner training (+/-) | Components: median (range) |
|----------------------------|--------------------------|------------------------------|----------------------------|
| Genome doubling            | +33 / -48                | +105 / -155                  | 6.0 (4-10)                 |
| APC                        | +39 / -37                | +124 / -118                  | 2.0 (1-4)                  |
| KRAS                       | +25 / -51                | +80 / -163                   | 4.0 (2-10)                 |
| TP53 [driver_mutation]     | +32 / -44                | +102 / -140                  | 2.0 (2-7)                  |
| MSI-H (MANTIS >0.4)        | +14 / -63                | +47 / -202                   | 6.0 (4-9)                  |
| MSI-H strict (MANTIS >0.6) | +11 / -63                | +36 / -202                   | 7.0 (3-10)                 |
| MYC                        | +15 / -48                | +50 / -156                   | 4.0 (2-5)                  |
| PI3K                       | +29 / -35                | +94 / -112                   | 4.0 (2-10)                 |
| TP53 [oncogenic_pathway]   | +41 / -23                | +132 / -73                   | 3.0 (1-6)                  |

### Repeat-to-repeat held-out prediction stability.

| Endpoint                   | Score Spearman rho | Class agreement |
|----------------------------|--------------------|-----------------|
| Genome doubling            | 0.850              | 0.857           |
| APC                        | 0.945              | 0.944           |
| KRAS                       | 0.825              | 0.858           |
| TP53 [driver_mutation]     | 0.948              | 0.931           |
| MSI-H (MANTIS >0.4)        | 0.856              | 0.940           |
| MSI-H strict (MANTIS >0.6) | 0.804              | 0.948           |
| MYC                        | 0.861              | 0.905           |
| PI3K                       | 0.721              | 0.787           |
| TP53 [oncogenic_pathway]   | 0.853              | 0.895           |

## Quality and provenance

- Patient-level aggregation: arithmetic mean across all supplied slide vectors.
- Maximum fraction outside endpoint-specific TCGA training ranges: 0.4%.
- TITAN input schema: 768 named features (`titan_000`–`titan_767`).
- Decomposition: seeded rSVD in fastPLS; binary classification: PLS-LDA.
- Tissue-source-site metadata: every row includes the grouped internal metric, random-minus-grouped change, analysed-site count, threshold-retention status, fold-size range and inner-site-separation audit.
- Binary reliability metadata: held-out average precision, cohort-specific PPV/NPV, outer and inner class-count minima, selected-component distribution and repeat-to-repeat prediction stability.
- Default model selection: binary models require at least 50 TCGA participants in each class; 17 smaller-class models are available only by explicit opt-in.
- Scope: TCGA tissue-source-site-grouped internal validation is not institutional or scanner-level external validation.
- Intended use: research-software demonstration only; TCGA benchmark without independent external validation.

## Endpoint interpretation and target sources

For continuous endpoints, the TCGA OOF prediction percentile (not probability) locates the prediction within repeated TCGA out-of-fold predictions and is not a clinical reference interval. For binary endpoints, every TCGA OOF score rank (not probability) is the relative position of an uncalibrated LDA score. It is not confidence or a clinical-risk estimate. PR-AUC is non-interpolated average precision for positive class 1. PPV and NPV are held-out estimates at the observed endpoint prevalence in this TCGA analysis and will change when prevalence changes. Rows marked exploratory/limited evidence are excluded from default inference and require explicit opt-in.

- **Genome doubling.** Binary whole-genome-doubling call derived from the tumour copy-number profile. *Analysed scale:* PLS-LDA class and uncalibrated discriminant score *Model-target source:* Taylor et al., Cancer Cell 2018, Table S2; doi:10.1016/j.ccell.2018.03.007.
- **APC.** Binary prediction of a qualifying protein-altering PASS mutation in APC; the displayed LDA score is not a probability. *Analysed scale:* PLS-LDA class and uncalibrated discriminant score *Model-target source:* MC3 consensus calls: Ellrott et al., Cell Systems 2018, doi:10.1016/j.cels.2018.03.002; cancer-gene eligibility: Bailey et al., Cell 2018, doi:10.1016/j.cell.2018.02.060.
- **KRAS.** Binary prediction of a qualifying protein-altering PASS mutation in KRAS; the displayed LDA score is not a probability. *Analysed scale:* PLS-LDA class and uncalibrated discriminant score *Model-target source:* MC3 consensus calls: Ellrott et al., Cell Systems 2018, doi:10.1016/j.cels.2018.03.002; cancer-gene eligibility: Bailey et al., Cell 2018, doi:10.1016/j.cell.2018.02.060.
- **TP53 [driver\_mutation].** Binary prediction of a qualifying protein-altering PASS mutation in TP53; the displayed LDA score is not a probability. *Analysed scale:* PLS-LDA class and uncalibrated discriminant score *Model-target source:* MC3 consensus calls: Ellrott et al., Cell Systems 2018, doi:10.1016/j.cels.2018.03.002; cancer-gene eligibility: Bailey et al., Cell 2018, doi:10.1016/j.cell.2018.02.060.
- **MSI-H (MANTIS >0.4).** Binary call indicating whether the source MANTIS score exceeds 0.4; the displayed LDA score is not a probability. *Analysed scale:* PLS-LDA class and uncalibrated discriminant score *Model-target source:* Bonneville et al., JCO Precision Oncology 2017; doi:10.1200/PO.17.00073.
- **MSI-H strict (MANTIS >0.6).** Sensitivity-analysis call: MANTIS >0.6 is positive, <0.4 is negative, and intermediate source values were excluded from training. *Analysed scale:* PLS-LDA class and uncalibrated discrim-

inant score *Model-target source*: Threshold documentation distributed with the cBioPortal TCGA PanCancer Atlas clinical sample files.

- **MYC**. Binary prediction that the MYC oncogenic pathway carries at least one qualifying genomic alteration; the displayed LDA score is not a probability. *Analysed scale*: PLS-LDA class and uncalibrated discriminant score *Model-target source*: Sanchez-Vega et al., Cell 2018, Table S4; doi:10.1016/j.cell.2018.03.035.
- **PI3K**. Binary prediction that the PI3K oncogenic pathway carries at least one qualifying genomic alteration; the displayed LDA score is not a probability. *Analysed scale*: PLS-LDA class and uncalibrated discriminant score *Model-target source*: Sanchez-Vega et al., Cell 2018, Table S4; doi:10.1016/j.cell.2018.03.035.
- **TP53 [oncogenic\_pathway]**. Binary prediction that the TP53 oncogenic pathway carries at least one qualifying genomic alteration; the displayed LDA score is not a probability. *Analysed scale*: PLS-LDA class and uncalibrated discriminant score *Model-target source*: Sanchez-Vega et al., Cell 2018, Table S4; doi:10.1016/j.cell.2018.03.035.
- **Aneuploidy score**. Number of chromosome arms with arm-level copy-number gain or loss; larger values indicate greater chromosomal imbalance. *Analysed scale*: source endpoint units *Model-target source*: Taylor et al., Cancer Cell 2018, Table S2; doi:10.1016/j.ccell.2018.03.007.
- **Deleted arm count**. Number of chromosome arms classified as deleted; larger values indicate a greater arm-level deletion burden. *Analysed scale*: source endpoint units *Model-target source*: Taylor et al., Cancer Cell 2018, Table S2; doi:10.1016/j.ccell.2018.03.007.
- **MANTIS score**. Genome-wide microsatellite-instability score from the MANTIS algorithm; larger values indicate stronger MSI evidence. *Analysed scale*: source endpoint units *Model-target source*: Kautto et al., Oncotarget 2017; doi:10.18632/oncotarget.20432. TCGA values were obtained from cBioPortal PanCancer Atlas records.
- **MSIsensor score**. Microsatellite-instability score from MSIsensor; larger values indicate a larger fraction of unstable microsatellite loci. *Analysed scale*: source endpoint units *Model-target source*: Niu et al., Bioinformatics 2014; doi:10.1093/bioinformatics/btt755. TCGA values were obtained from cBioPortal PanCancer Atlas records.
- **Aneuploidy Score**. Aneuploidy burden distributed with the TCGA immune landscape resource; larger values indicate more arm-level copy-number alterations. *Analysed scale*: source endpoint units *Model-target source*: Thorsson et al., Immunity 2018, PanImmune\_MS; doi:10.1016/j.immuni.2018.03.023.
- **Lymphocyte Infiltration Signature Score**. Gene-expression signature summarising lymphocyte-associated transcriptional infiltration; it is not a direct cell count. *Analysed scale*: source endpoint units *Model-target source*: Thorsson et al., Immunity 2018, PanImmune\_MS; doi:10.1016/j.immuni.2018.03.023.
- **Nonsilent Mutation Rate**. Rate of coding mutations that alter the encoded protein. TITANPred reports the model output on the analysis  $\log(1+x)$  scale. *Analysed scale*:  $\log_{10}$ -transformed source endpoint units *Model-target source*: Thorsson et al., Immunity 2018, PanImmune\_MS; doi:10.1016/j.immuni.2018.03.023.
- **SNV Neoantigens**. Predicted neoantigen burden arising from single-nucleotide variants. TITANPred reports the model output on the analysis  $\log(1+x)$  scale. *Analysed scale*:  $\log_{10}$ -transformed source endpoint units *Model-target source*: Thorsson et al., Immunity 2018, PanImmune\_MS; doi:10.1016/j.immuni.2018.03.023.
- **Silent Mutation Rate**. Rate of coding mutations that do not change the encoded amino acid. TITANPred reports the model output on the analysis  $\log(1+x)$  scale. *Analysed scale*:  $\log_{10}$ -transformed source endpoint units *Model-target source*: Thorsson et al., Immunity 2018, PanImmune\_MS; doi:10.1016/j.immuni.2018.03.023.
- **TIL Regional Fraction**. Estimated fraction of tumour image regions containing tumour-infiltrating lymphocytes in the TCGA immune landscape resource. *Analysed scale*: source endpoint units *Model-target source*: Thorsson et al., Immunity 2018, PanImmune\_MS; doi:10.1016/j.immuni.2018.03.023.